

Supporting Information: Absorption *versus* Adsorption: High-Throughput Computation of Impurities in 2D Materials

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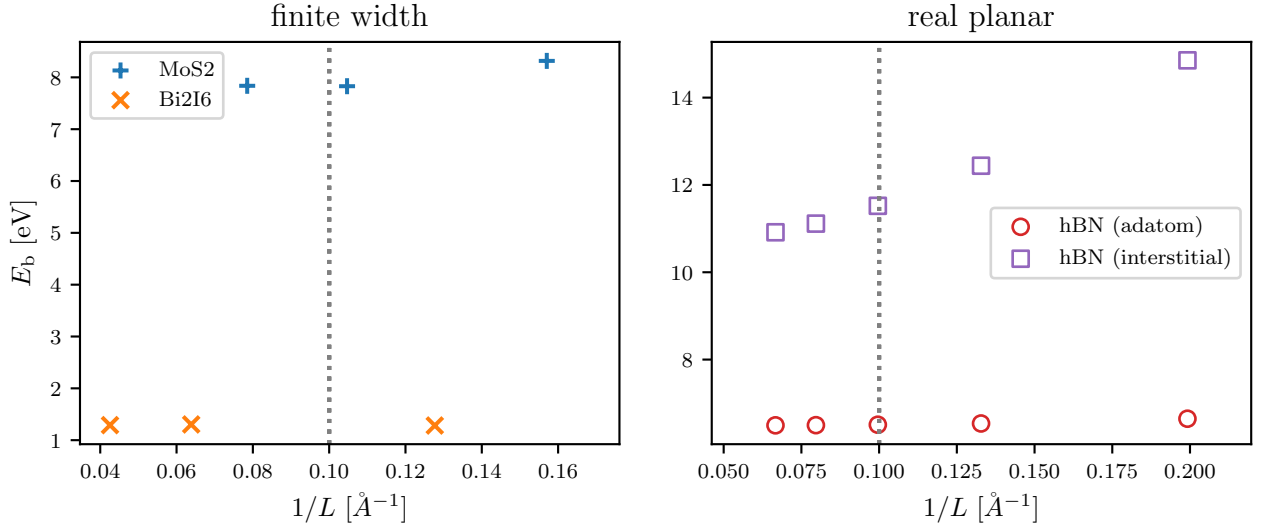
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Supplementary Note 1: Supercell Size Convergence

Before launching the study on the entire dataset, we conducted a convergence analysis with respect to in plane supercell sizes. The outcomes of this benchmark are shown in Supplementary Figure 1: in the left panel, we see the binding energy of an interstitial defects in MoS₂ and Bi₂I₆, respectively. One can see that the binding energy varies on the order of several hundreds of meV for the smallest supercell in MoS₂, but once the supercell sizes are larger than 10 Å (*i.e.* $\frac{1}{L} < 0.1 \text{ Å}^{-1}$), the binding energy values are well converged. Bi₂I₆, a system with more atoms in the primitive unit cell compared to MoS₂, converges for even larger supercell sizes. Based on these two observations we choose the minimum supercell size for the entire study to be at least 10 Å. On the right hand side of Supplementary Figure 1 we compare the convergence behaviour for an adatom and interstitial in the real planar host structure of hBN. The adatom is well converged already for small supercell sizes (which is intuitive as we do not expect adatoms to introduce big structural changes to the host crystal). The interstitial position, however, is not even converged for supecell sizes well above 10 Å. Therefore, we excluded real planar systems like hBN and graphene from our full study.



Supplementary Figure 1: **Convergence behaviour of interstitial and adsorption sites in terms of supercell size.** Left: binding energy of an interstitial atom in MoS₂ and Bi₂I₆ (2D materials with finite out of plane width) as a function of the inverse supercell size $1/L$ where $L = \frac{L_x + L_y}{2}$. Right: binding energy of an interstitial defect and atom in hBN (real planar system with no out of plane width) as a function of the inverse supercell size $\frac{1}{L}$.

Supplementary Note 2: Overview of host materials

Supplementary Table 1: **List of host systems.** Overview of the 53 host crystals with host formula, space group, pristine band gap, magnetic information. Only Cr₂I₆ is a magnetic monolayer, all other host systems are non-magnetic.

Host formula	Space group	Band Gap
P ₄	53	0.90 eV
Ga ₂ Se ₂	123	0.00 e
MoSe ₂	187	1.32 eV
W ₂ Te ₄	11	0.00 eV
ZrS ₂	164	1.16 eV
Bi ₂ I ₆	147	1.38 eV
Cr ₂ I ₆	162	0.89 eV
Hf ₂ Te ₆	59	0.00 eV

PbI ₂	164	1.50 eV
Nb ₄ C ₃	164	0.00 eV
NiSe ₂	164	0.06 eV
PtS ₂	164	1.69 eV
WTe ₂	187	0.73 eV
Si ₂	164	0.00 eV
Ti ₃ C ₂ H ₂ O ₂	187	0.00 eV
Ga ₂ Te ₂	187	1.30 eV
SnS ₂	164	1.59 eV
TaS ₂	164	0.00 eV
Mo ₂ CO ₂	164	0.00 eV
Sn ₂	164	0.06 eV
In ₂ Se ₂	123	0.00 eV
MoTe ₂	187	0.93 eV
NbSe ₂	187	0.00 eV
SnS ₂	187	0.76 eV
TiO ₂	164	2.70 eV
ZrSe ₂	164	0.34 eV
Ge ₂ H ₂	164	0.92 eV
Pd ₂ Se ₄	14	1.31 eV
SnSe ₂	164	0.76 eV
WS ₂	187	1.53 eV
Nb ₂ CO ₂	164	0.00 eV
GaN	187	1.88 eV
MoS ₂	164	0.00 eV
PtSe ₂	164	1.17 eV
Re ₄ S ₈	2	1.27 eV

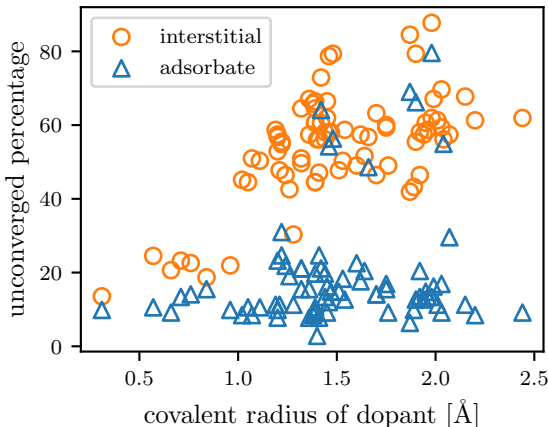
MoS ₂	187	1.58 eV
BiITe	156	0.41 eV
As ₂	164	1.48 eV
C ₂ H ₂	164	3.46 eV
TiS ₂	164	0.02 eV
W ₂ Se ₄	11	0.03 eV
MoSSe	156	1.45 eV
HfS ₂	164	1.22 eV
TaSe ₂	164	0.00 eV
Ti ₂ CO ₂	164	0.32 eV
Re ₄ Se ₈	2	1.12 eV
WSe ₂	187	1.24 eV
V ₂ CO ₂	164	0.00 eV
C ₆ N ₂	191	0.40 eV
Ti ₂ S ₆	11	0.29 eV
Ge ₂	164	0.03 eV
HfSe ₂	164	0.43 eV
NbS ₂	187	0.00 eV

Supplementary Note 3: Chemical Potentials

To evaluate the defect formation energy we set to the chemical potential of the dopant element to the energy of the crystalline phases.¹ The elemental crystal structures were relaxed using a workflow similar to the one described in the main text, but with an initial volume relaxation step. To ensure that the plane wave basis is accurate even when the volume is changed, the plane wave energy cutoff is increased to 700 eV and kinetic energy cutoff to 1400 eV. The k -point grid is set up automatically with *htrk* using the Monkhorst-Pack method.² Electronic tolerance, ionic tolerance, and FFT grid are set to 10^{-6} eV, $5 \cdot 10^{-5}$ eV, and twice the largest wavevector, respectively. The last step is restarted until the energy difference between two steps is less than $5 \cdot 10^{-4}$ eV. The formation energies are stored in the database.

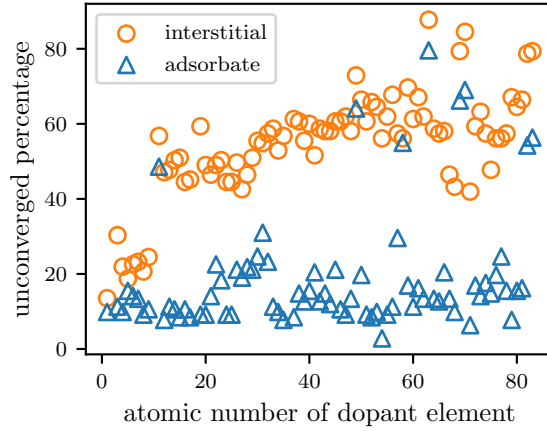
Supplementary Note 4: Convergence statistics

Since a number of calculations did not converge within the set of ionic steps, it is of interest to check whether such systems could generally be assumed to be unphysical/unstable. For example, systems with a dopant placed in a small interstitial volume is unlikely to result in a stable configuration and is simultaneously expected to require many relaxation steps to reach a local minimum. If such a correlation exists, unconverged calculations would be less of a concern, as they would be unlikely to produce physically relevant information even if they were brought to convergence. When considering all the calculations that did not converge in their ionic relaxation (in the following referred to as 'unconverged') we see that 36% of the interstitial positions do not converge while only 20% of the set up adsorption sites do not converge.



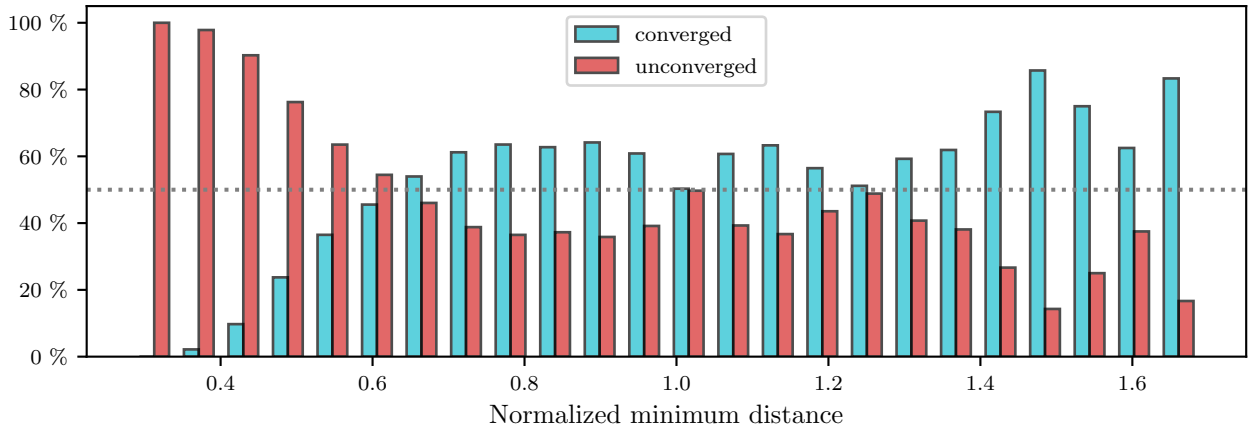
Supplementary Figure 2: **Calculation of unconverged percentage and covalent radii.** Percentage of unconverged calculations as a function of covalent radius of the dopant element for adsorption sites (blue triangles) and interstitials (orange circles).

Supplementary Figure 2 shows the percentage of unconverged calculations as a function of the covalent radius of the dopant atom across all host crystals for interstitial and adsorption positions, respectively. It is clear that the rate of unconverged systems is higher for interstitial doping. On one hand, the convergence rate for interstitials shows a clear correlation with the covalent radius of the dopant atom. In general, it is difficult to converge calculations for



Supplementary Figure 3: **Calculation of unconverged percentage and dopant atomic numbers.** Percentage of unconverged calculations as a function of atomic number of the dopant element for adsorption sites (blue triangles) and interstitials (orange circles).

interstitial dopants with covalent radius above 1 Å. No such correlation is seen for dopants at adsorption sites, although there are a few outliers with above average unconverged rates.



Supplementary Figure 4: **Convergence behaviour and nearest neighbor distances for interstitial sites.** Bar diagram of the percentage of converged vs. unconverged calculations as a function of the normalized minimum distance parameter $D_{\min}^{\text{norm}}(\text{I, A})$. Each pair of bins adds up to 100 %.

In Supplementary Figure 3, a similar trend is obtained in terms of the unconverged percentage as a function of dopant type and atomic number of dopant element: the convergence for systems doped at adsorption sites is generally better and not strongly correlated with

the atomic number of the dopant element. For interstitials, however, a low unconverged rate is only present for dopant elements with atomic numbers lower than 15, whereas heavier dopant elements increase the percentage of unconverged calculations.

Interstitial sites seem to be the bottleneck, and an in-depth analysis of those systems is needed. For a given system doped with an interstitial atom I, there exists one atom A in the host crystal that possesses the minimum distance $D_{\min}(\text{I}, \text{A})$ to the interstitial site. To quantify the convergence behavior of the interstitial systems, we define a normalized minimum distance parameter as

$$D_{\min}^{\text{norm}}(\text{I}, \text{A}) = \frac{D_{\min}(\text{I}, \text{A})}{r_{\text{C}}(\text{I}) + r_{\text{C}}(\text{A})}. \quad (1)$$

Here, $r_{\text{C}}(\text{A})$ and $r_{\text{C}}(\text{I})$ denote the covalent radii for the corresponding host atom and interstitial atom, respectively. Supplementary Figure 4 visualizes the percentage of converged and unconverged calculations for interstitial systems as a function of $D_{\min}^{\text{norm}}(\text{I}, \text{A})$. For low normalized minimum distances almost no calculations converge but with increasing $D_{\min}^{\text{norm}}(\text{I}, \text{A})$ the percentage of converged calculations increases. For normalized minimum distance values above 0.6, the probability of converging an interstitial position is generally larger than 50%. This trend is confirmed by the cumulative percentage of converged interstitial calculations: 56% for $D_{\min}^{\text{norm}}(\text{I}, \text{A}) > 0.6$ and only 25% for $D_{\min}^{\text{norm}}(\text{I}, \text{A}) < 0.6$.

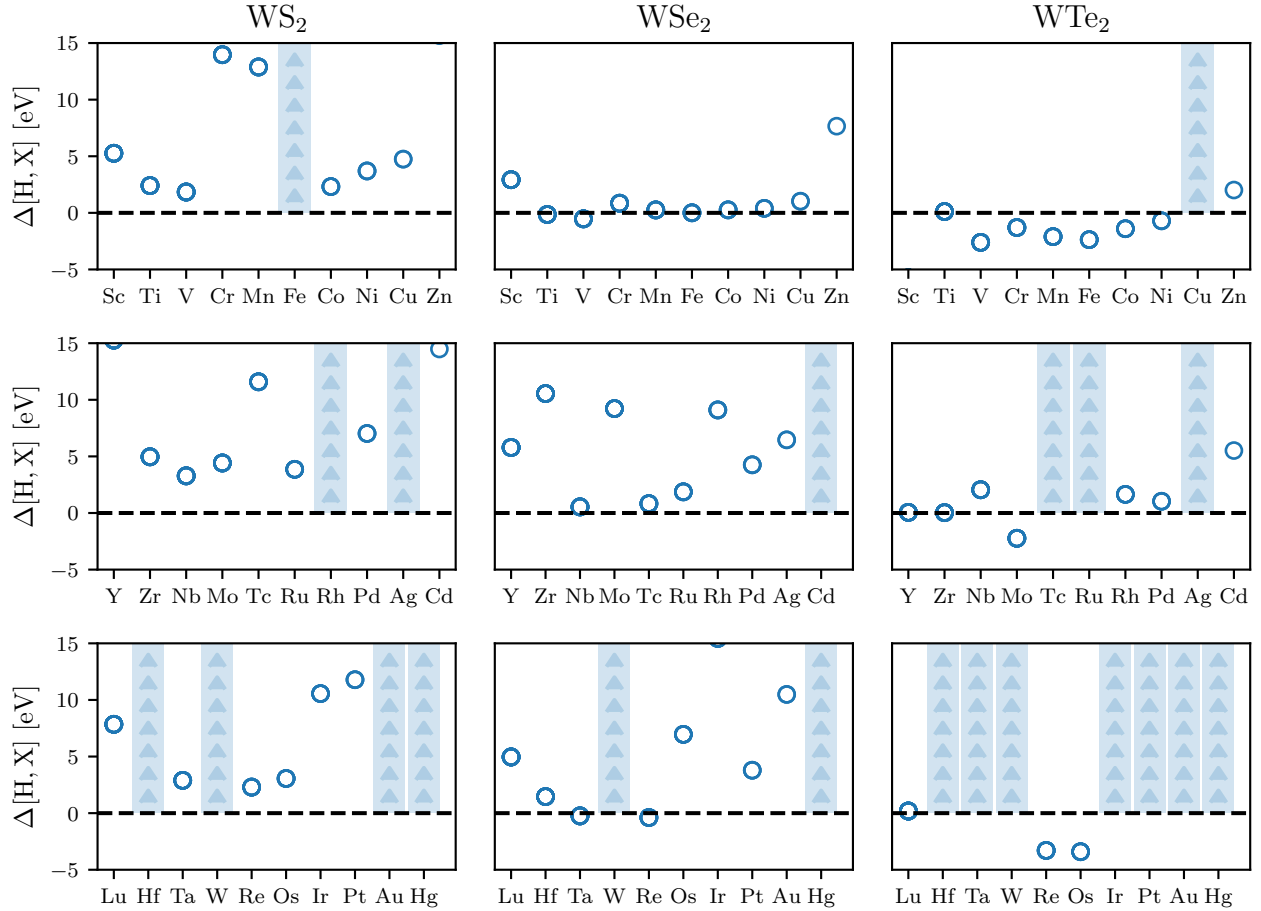
We note that the ionic convergence settings are set relatively low due to the vast number of systems that needed to be calculated. In particular, not to waste computational resources, the ionic steps are limited to 20 since the ions may oscillate back and forth for some systems. Some unconverged structures can be used as a good starting point for a detailed subsequent analysis. Unconverged systems can be found in the database *via* the query: `converged=False`.

In summary, we saw that our unconverged calculations can give insight into the stability of adsorption and interstitial site doping of 2D materials. For interstitial defects, the overall

large percentage of unconverged calculations stems from unphysical systems where an interstitial site simply does not have enough space inside the monolayer. For instance, we saw that dopant atoms with large covalent radii are particularly challenging systems. Therefore, in the context of high-throughput calculations, it is important to choose physically reasonable initial configurations for interstitial positions (we suggest normalized minimum distances of at least 0.6) to end up with convergence percentages of above 50%. On top of the structural challenges, our approach of doping the monolayers with most of the elements from the periodic table regardless of the chemical compatibility leads to large percentages of unconverged systems. In a future study, a thorough analysis of the convergence behavior in terms of a chemical compatibility descriptor might be helpful for more insight. The public availability of the dataset with its documentation online should make such an additional study easy to conduct.

The overall low percentage of converged structures is mostly due to the challenging task of relaxing interstitial defects. We note that a more computationally heavy workflow that allows for a larger number of ionic relaxation steps might improve that number in future studies. Many of the generated initial structures turned out to be unphysical systems, and we suggest to always choose a reasonable normalized minimum distance when setting up interstitial structures. Furthermore, the amount of unconverged calculations is correlated with the atomic species a given system is doped with, in particular for interstitials.

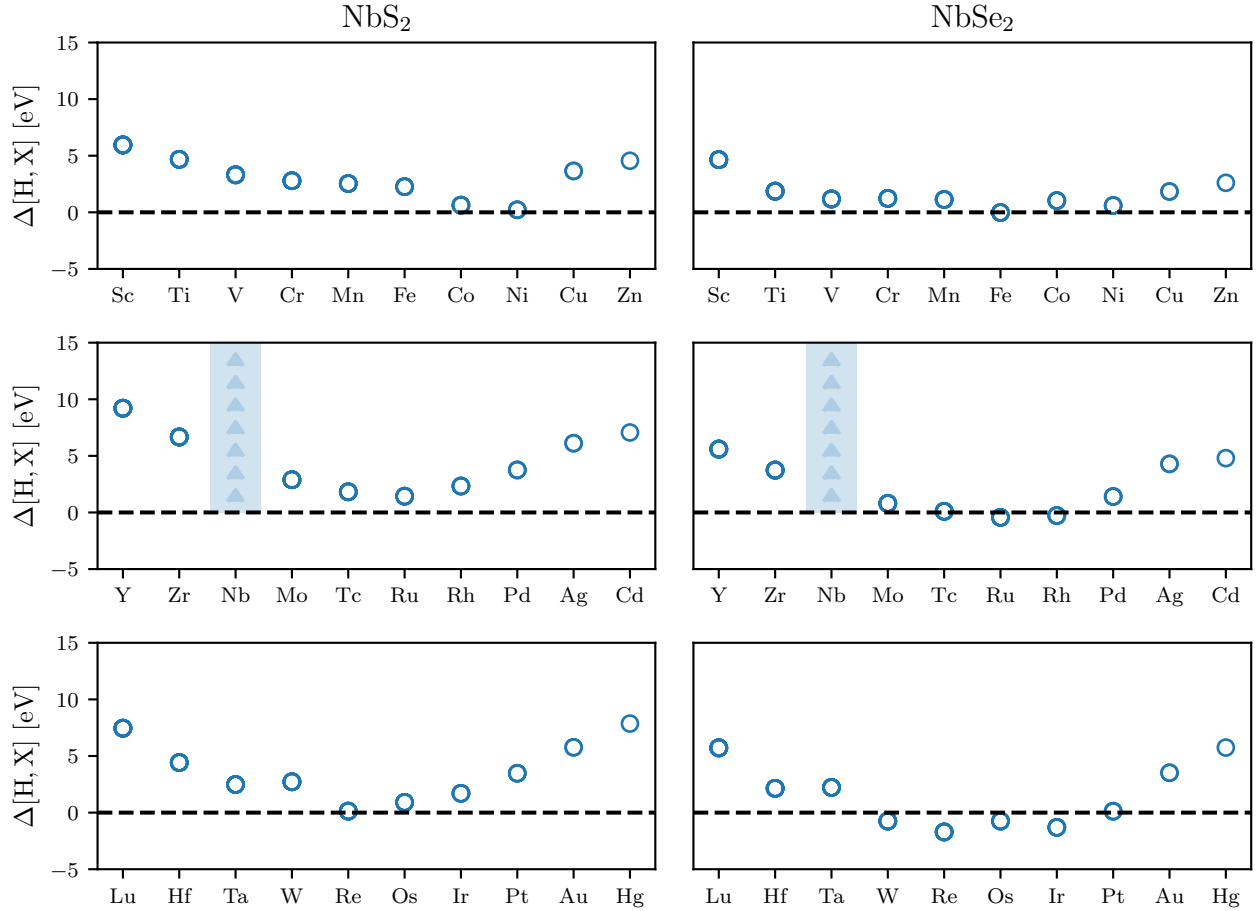
Supplementary Note 5: TM-doping for 2H-MX₂ Monolayers



Supplementary Figure 5: **Transition metal doping in transition metal dichalcogenides 2H-WX₂.** $\Delta[H, X]$ values for transition metal doped WS₂ (left column), WSe₂ (middle column) and WTe₂ (right column) as a function of the doping element. Blue bars represent systems where we were not able to converge both adsorption sites and interstitial positions.

In addition to the interstitial and adsorption site trends observed for MoS₂, MoSe₂, MoTe₂, we conducted a similar analysis for the WX₂ (X=S,Se,Te) monolayers. The results are shown in Supplementary Figure 5: just like for the case of MoX₂ (see main manuscript), interstitial sites are not energetically favorable in the monolayer with the smallest in plane lattice constant, *i.e.* WS₂. However, when going to the larger unit cells, interstitials become

close in energy for WSe_2 and even favorable for the system with the largest lattice constant, *i.e.* WTe_2 . The same trend is also observed for NbS_2 and NbSe_2 (Supplementary Figure 6).



Supplementary Figure 6: **Transition metal doping in transition metal dichalcogenides 2H-NbX_2 .** $\Delta[\text{H}, \text{X}]$ values for transition metal doped NbS_2 (left column), NbSe_2 (middle column) and NbTe_2 (right column) as a function of the doping element. Blue bars represent systems where we were not able to converge both adsorption sites and interstitial positions.

Supplementary Note 6: Bonding of Impurities

To illustrate how diverse the different systems are and the bonding of the impurities, we look more closely at three dopants B, C, and N in MoS₂. Table 2 shows $\Delta [\text{MoS}_2, \text{X}]$ as well as the lowest energy interstitial and adsorbate sites. Based on the $\Delta [\text{H}, \text{X}]$, B and C prefer the interstitial site, whereas N prefers the adsorbate site.

Supplementary Table 2: **B, C, and N impurities in MoS₂**. Preferred sites are bolded, and the multiple sites (int0, int3, int4) relaxed to the same position and thus have the same lowest energy.

Dopant	$\Delta [\text{MoS}_2, \text{X}]$ (eV)	Int site	Abs site
B	-0.85	ads2*	ads1
C	-0.80	int0, int3, int4	ads1
N	0.55	int0, int3, int4	ads0

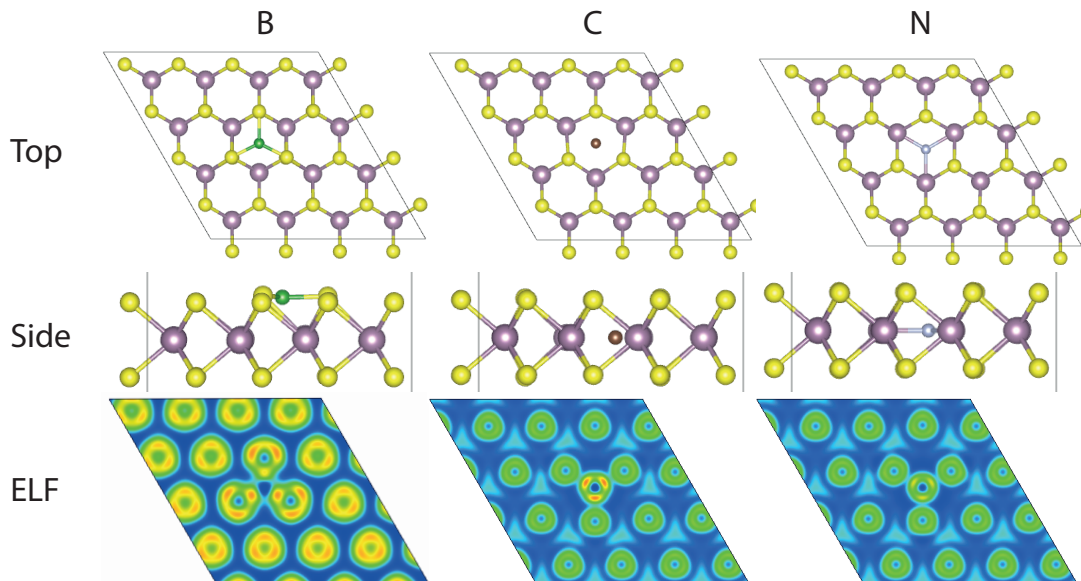
*started as an adsorbate site but moved into the host and became an interstitial site.

The different structures for the interstitial sites are displayed in Supplementary Figure 7, and for adsorbate sites in Supplementary Figure 8. To better show the bonds, we made electron localization function (ELF) plots for the different systems where large values between atoms indicate bonding.

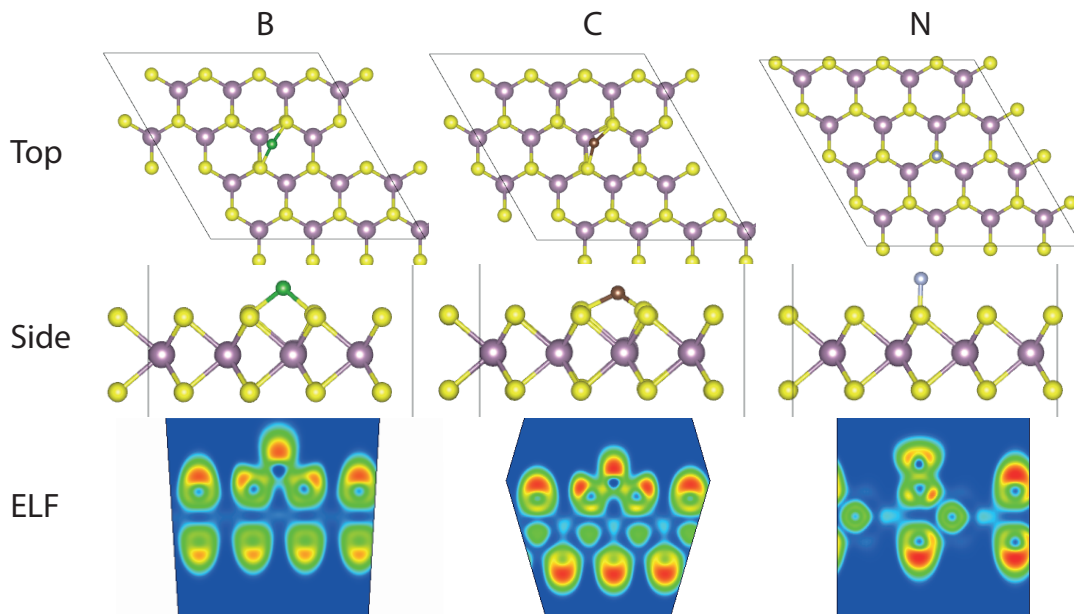
Supplementary Figure 7 shows that the B and C interstitials strongly bond to the S and Mo, respectively. However, the N interstitial is weakly bonded to the Mo compared to the C interstitial.

Supplementary Figure 8 shows that the B and C adsorbate sits between two S and weakly bonds to both (ads1 site), whereas the N adsorbate sits directly on top of one S atom (ads0 site). The energy difference for the N adsorbate to sit at the ads0 site compared to the ads1 site is 1.19 eV.

These examples show the diverse possibilities that can emerge from changing the impurity group in the periodic table. Not only are the different bonding physics but also at different positions. It highlights the importance of testing several different positions.



Supplementary Figure 7: **B, C, N lowest energy interstitials in MoS_2** . The interstitial is shown from a top and side view. The 2D ELF plane is plotted for the three nearest S neighbors for the B interstitial and for the three nearest Mo neighbors for the C and N interstitials. Made using VESTA.³



Supplementary Figure 8: **B, C, N lowest energy adsorbates in MoS_2** . The adsorbate is shown from a top and side view. The 2D ELF plane is plotted for the two nearest S neighbors and B or C impurity, and for the nearest S neighbor, N impurity, and second nearest Mo neighbor. Made using VESTA.³

Supplementary References

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